

GitLab - 使用生成式 AI 自動執行程式碼審查作業

來源：<https://codelabs.developers.google.com/genai-for-dev-gitlab-code-review?hl=zh-tw>

步驟數：13

GitLab - 運用生成式 AI 自動化程式碼審查

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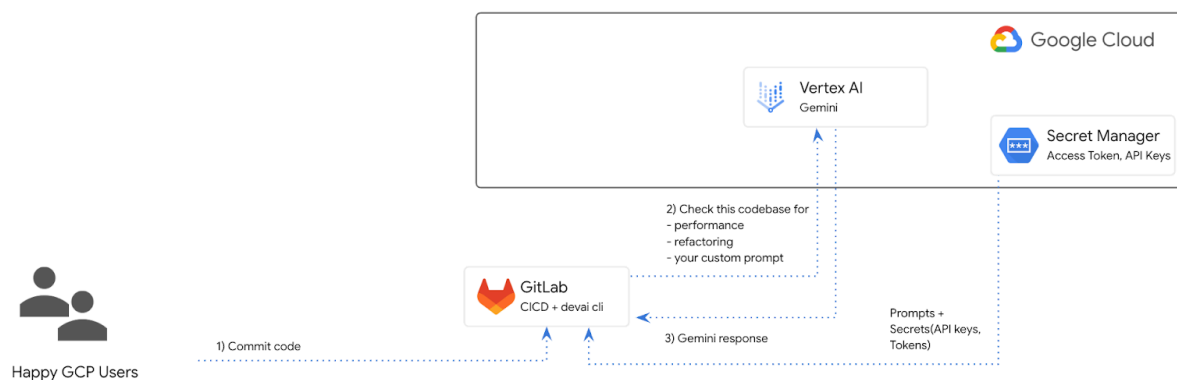
步驟 1 · 約 0 分鐘

1. 總覽

在本實驗室中，您將設定 GitLab CI/CD 管道，並與 Gemini 整合，自動執行程式碼審查步驟。

Code review automation with GenAI

Gemini + GitLab + JIRA



學習目標

在本實驗室中，您將瞭解如何執行下列操作：

- 如何在 GitLab 中新增生成式 AI 程式碼審查自動化步驟
- 如何在本機執行 **devai** CLI，自動執行程式碼審查

必要條件

- 本實驗室假設您已熟悉 Cloud 控制台和 Cloud Shell 環境。

步驟 2 · 約 0 分鐘

2. 設定和需求

設定 Cloud 專案

1. 登入 [Google Cloud 控制台](#)，然後建立新專案或重複使用現有專案。如果沒有 Gmail 或 Google Workspace 帳戶，請先[建立帳戶](#)。

The screenshot shows the Google Cloud console interface. At the top, there is a 'Select a project' dropdown menu. Below it, the 'Select a project' page is visible, featuring a search bar labeled 'Search projects and folders'. To the right of the search bar is a 'NEW PROJECT' button. Below the search bar, the 'New Project' form is shown. It includes a 'Project name' field with the value 'My Project 13475', a 'Project ID' field with the value 'inbound-analogy-389614' (highlighted with a blue box), and a 'Location' field. The 'Project ID' field has a note: 'It cannot be changed later. EDIT'. At the bottom of the form are 'CREATE' and 'CANCEL' buttons.

- **專案名稱**是這個專案參與者的顯示名稱。這是 Google API 未使用的字元字串。你隨時可以更新。
- **專案 ID** 在所有 Google Cloud 專案中都是不重複的，而且設定後即無法變更。Cloud 控制台會自動產生專屬字串，通常您不需要在意該字串為何。在大多數程式碼研究室中，您需要參照專案 ID (通常標示為 `PROJECT_ID`)。如果您不喜歡產生的 ID，可以產生另一個隨機 ID。你也可以嘗試使用自己的名稱，看看是否可用。完成這個步驟後就無法變更，且專案期間會維持不變。
- 請注意，有些 API 會使用第三個值，也就是「專案編號」。如要進一步瞭解這三種值，請參閱[說明文件](#)。

注意：專案 ID 在全域中是唯一的，選定後就無法再供他人使用。您是該 ID 的唯一使用者。即使專案已刪除，ID 也無法再次使用

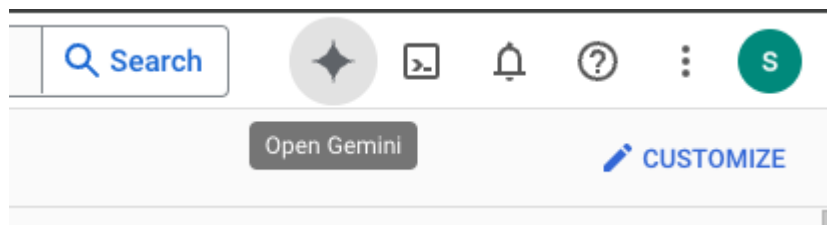
注意：如果使用 Gmail 帳戶，可以將預設位置設為「沒有機構」。如果你使用 Google Workspace 帳戶，請選擇適合貴機構的位置。

2. 接著，您需要在 Cloud 控制台中[啟用帳單](#)，才能使用 Cloud 資源/API。完成這個程式碼研究室的費用不高，甚至可能完全免費。如要關閉資源，避免在本教學課程結束後繼續產生費用，請刪除您建立的資源或專案。Google Cloud 新使

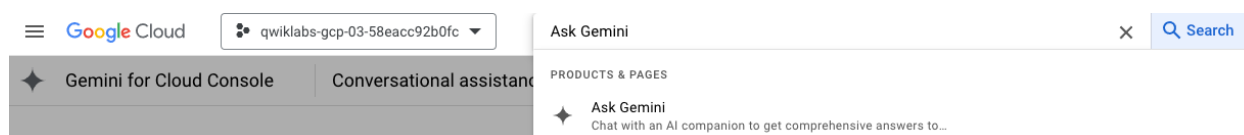
用戶可參加[價值\\$300 美元的免費試用計畫](#)。

環境設定

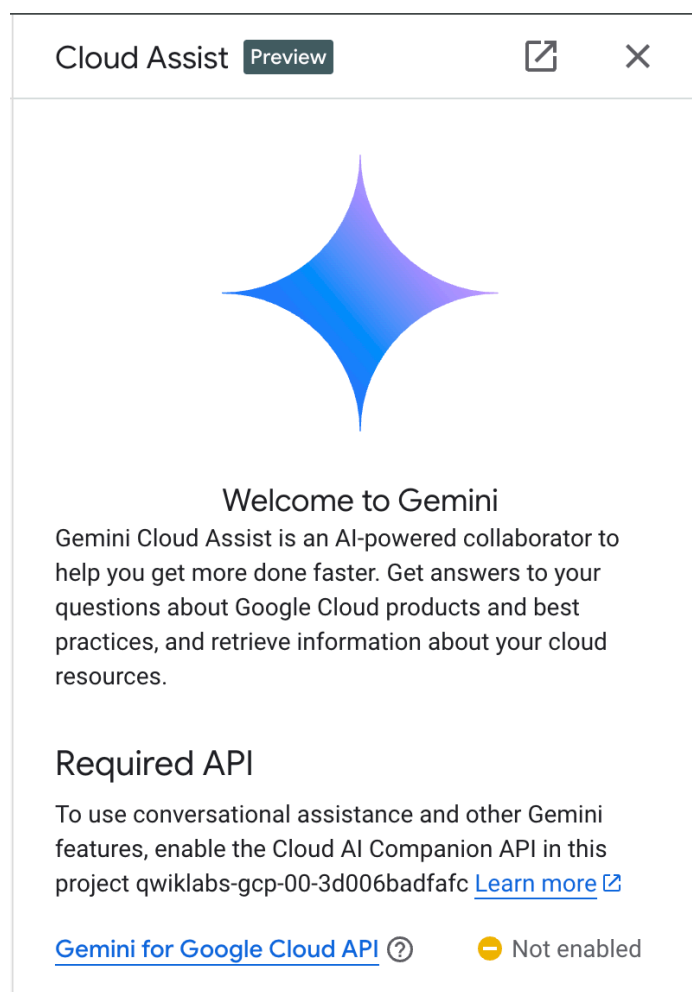
開啟 Gemini 對話。



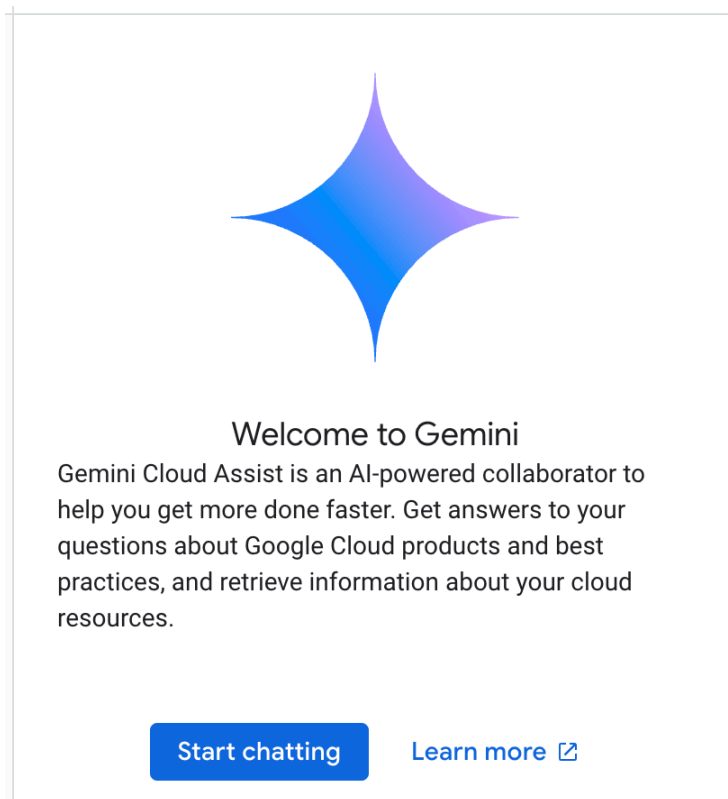
或在搜尋列中輸入「問問 Gemini」。



啟用 Gemini for Google Cloud API：



按一下「`Start chatting`」，然後按照其中一個範例問題操作，或自行輸入提示詞來試用。



建議提示詞：

- 請用 5 個重點說明 Cloud Run。
- 您是 Google Cloud Run 產品經理，請用 5 個簡短重點向學生說明 Cloud Run。
- 您是 Google Cloud Run 產品經理，請用 5 個簡短重點向認證 Kubernetes 開發人員說明 Cloud Run。
- 您是 Google Cloud Run 產品經理，請以 5 個簡短重點，向資深開發人員說明何時該使用 Cloud Run，何時該使用 GKE。

若要進一步瞭解如何撰寫更優質的提示，請參閱[提示指南](#)。

瞭解 Gemini for Google Cloud 如何使用您的資料

Google 對隱私權的承諾

Google 是業界首批發布 [AI/機器學習隱私權承諾](#) 的公司，該文提到我們的信念：除了極致的安全性之外，客戶也應該對儲存在雲端的自家資料保有最大的掌控權。

您提交及收到的資料

您向 Gemini 提出的問題 (包括提交給 Gemini 分析或完成的任何輸入資訊或程式碼)，都稱為提示。您從 Gemini 取得的答案或程式碼完成建議稱為「回覆」。我們不會將您的提示詞或 Gemini 的回覆用做模型訓練資料。

提示加密

當您將提示提交給 Gemini 時，您的資料會在傳輸過程中加密，然後輸入 Gemini 的基礎模型。

Gemini 生成的節目資料

Gemini 是以 Google Cloud 第一方程式碼和精選第三方程式碼訓練而成。您必須對程式碼的安全性、測試和效用負責，包括 Gemini 提供的任何程式碼完成、生成或分析功能。

[進一步瞭解](#) Google 如何處理提示。

3. 測試提示的選項

如要變更/延長現有的 devai CLI 提示，有幾種做法可供選擇。

- [Vertex AI Studio](#)

Vertex AI Studio 是 Google Cloud Vertex AI 平台的一部分，專門用於簡化及加速生成式 AI 模型的開發和使用。

- [Google AI Studio](#)

Google AI Studio 是網頁工具，可讓您設計提示工程原型，並透過 Gemini API 進行實驗。

- [Gemini 網頁應用程式](#) (gemini.google.com)

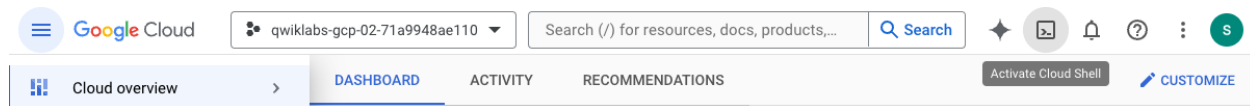
Google Gemini 網頁應用程式 (gemini.google.com) 是一款網頁工具，可協助您探索及運用 Google Gemini AI 模型強大的功能。

- [Android](#)和 [iOS 版 Google 應用程式](#)
-

步驟 4 · 約 0 分鐘

4. 建立服務帳戶

按一下搜尋列右側的圖示，啟用 Cloud Shell。



在開啟的終端機中，啟用使用 Vertex AI API 和 Gemini 對話 所需的服務。

```
gcloud services enable \  
  aiplatform.googleapis.com \  
  cloudaiaccompanion.googleapis.com \  
  cloudresourcemanager.googleapis.com \  
  secretmanager.googleapis.com
```

如果系統提示您授權，請點選「授權」繼續操作。

Authorize Cloud Shell

gcloud is requesting your credentials to make a GCP API call.

Click to authorize this and future calls that require your credentials.

REJECT

AUTHORIZE

執行下列指令，建立新的服務帳戶和金鑰。

您將使用這個服務帳戶，透過 CICD 管道對 Vertex AI Gemini API 發出 API 呼叫。

```
PROJECT_ID=$(gcloud config get-value project)  
SERVICE_ACCOUNT_NAME='vertex-client'  
DISPLAY_NAME='Vertex Client'  
KEY_FILE_NAME='vertex-client-key'  
  
gcloud iam service-accounts create $SERVICE_ACCOUNT_NAME --display-name "$DISPLAY_NAME"
```

授予角色。

```
gcloud projects add-iam-policy-binding $PROJECT_ID --  
member="serviceAccount:$SERVICE_ACCOUNT_NAME@$PROJECT_ID.iam.gserviceaccount.com" --  
role="roles/aiplatform.admin" --condition None
```

```
gcloud projects add-iam-policy-binding $PROJECT_ID --  
member="serviceAccount:$SERVICE_ACCOUNT_NAME@$PROJECT_ID.iam.gserviceaccount.com" --  
role="roles/secretmanager.secretAccessor" --condition None
```

```
gcloud iam service-accounts keys create $KEY_FILE_NAME.json --iam-  
account=$SERVICE_ACCOUNT_NAME@$PROJECT_ID.iam.gserviceaccount.com
```

步驟 5 · 約 0 分鐘

5. 將 GitHub 存放區匯入 GitLab 存放區

前往 <https://gitlab.com/projects/new>，然後選取「Import project」/「Repository by URL」選項：

Git 存放區網址：

```
https://github.com/GoogleCloudPlatform/genai-for-developers.git
```

在「專案網址」下方，選取你的 GitLab 使用者 ID

將「瀏覽權限」設為 **Public**。

按一下「Create Project」即可開始匯入程序。

如果看到 GitHub 存放區網址無效的錯誤訊息，請[建立新的 GitHub 權杖](#)(精細)，並提供公開存放區的唯讀存取權，然後再次嘗試匯入，並提供 GitHub 使用者 ID 和權杖。

步驟 6 · 約 0 分鐘

6. 新增 GitLab CI/CD 管道變數

接下來，您要啟用 GitLab CI/CD 管道，以便在變更推送至存放區時執行程式碼審查。

在瀏覽器中開啟 GitLab 存放區，然後前往「Settings / CI/CD」部分。

展開 `Variables` 部分，然後按一下「Add variable」。

新增變數時，請務必取消勾選所有核取方塊。

範例：

Add variable

Type

Variable (default)

Environments

All (default)

Flags

☐ Protect variable

Export variable to pipelines running on protected branches and tags only.

☐ Mask variable

Variable will be masked in job logs. Requires values to meet regular expression requirements.

☐ Expand variable reference

\$ will be treated as the start of a reference to another variable.

Description (optional)

The description of the variable's value or usage.

Key

GITLAB_BASE_BRANCH

You can use CI/CD variables with the same name in different places, but the variables might overwrite each other. [What is the order of precedence for variables?](#)

Value

main

Variable value will be evaluated as raw string.

Add variable

Cancel

新增 3 個變數：

- `PROJECT_ID`：您的 Qwiklabs 專案 ID
- `LOCATION` - us-central1
- `GOOGLE_CLOUD_CREDENTIALS`

針對 `GOOGLE_CLOUD_CREDENTIALS` 變數值，請使用上方章節中建立的服務帳戶金鑰。在 Google Cloud Shell 中執行這項指令，然後複製/貼上該值。

```
cat ~/vertex-client-key.json
```

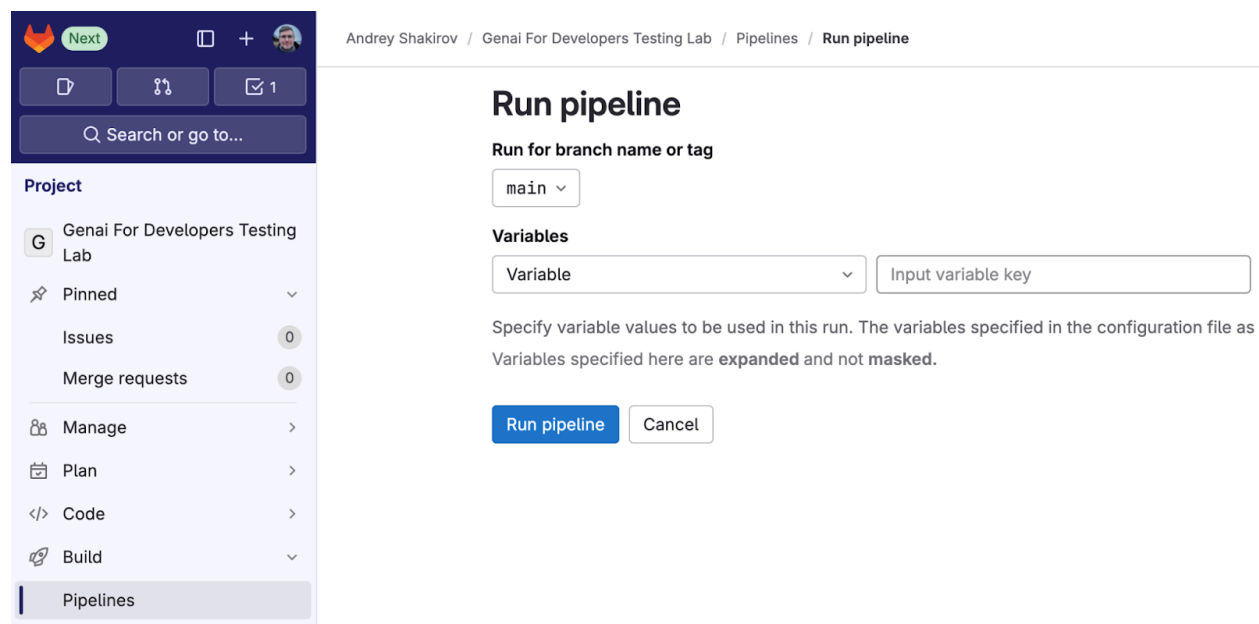
CI/CD 變數檢視畫面：

CI/CD Variables </> 3				Reveal values	Add variable
Key ↑	Value	Environments	Actions		
GOOGLE_CLOUD_CREDENTIALS	*****	All (default)			
LOCATION	*****	All (default)			
PROJECT_ID	*****	All (default)			

步驟 7 · 約 0 分鐘

7. 執行 GitLab CI/CD 管道

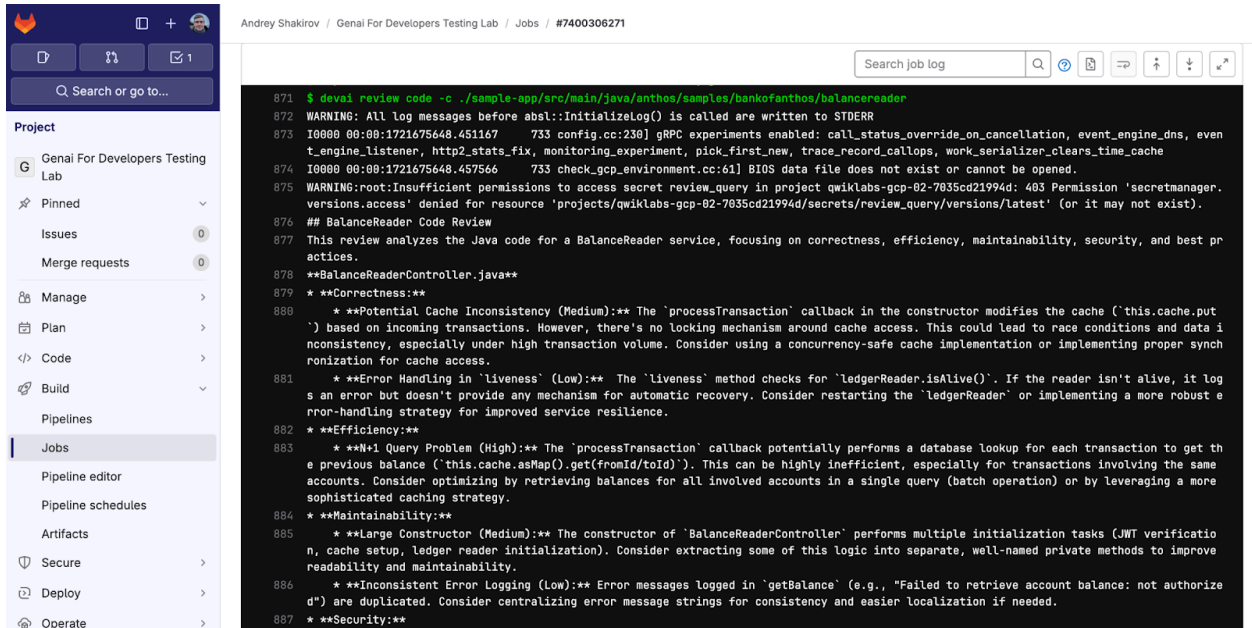
在 GitLab UI 中開啟「Build / Pipelines」，然後按一下「Run Pipeline」。



步驟 8 · 約 0 分鐘

8. 查看 GitLab 管道輸出內容

在 GitLab UI 中開啟「Build / Jobs」，並檢查管道輸出內容。



管道指令總覽

GitLab 管道設定：[.gitlab-ci.yml](#)

測試涵蓋範圍指令：

```
devai review testcoverage -c ./sample-app/src
```

這項 `devai review testcoverage` 指令會使用生成式 AI 模型 Gemini，分析程式碼和相關聯的測試套件 (如有)。這項工具會評估所提供程式碼的測試涵蓋範圍，找出有單元測試和沒有單元測試的檔案和方法。然後，這項指令會運用模型提供涵蓋範圍摘要，包括涵蓋的程式碼行和分支/條件等指標。根據分析結果，這項工具會提供建議，協助您改善測試涵蓋範圍、建議要新增的特定測試，以及提供測試最佳做法的一般建議。最後，透過指令列輸出 Gemini 模型的相關回覆，包括缺少涵蓋範圍的檔案，以及改善現有測試的建議。

程式碼審查指令：

```
devai review code -c ./sample-
app/src/main/java/anthos/samples/bankofanthos/balancereader
```

`devai review code` 指令會使用生成式 AI 模型，對提供的程式碼片段進行全面程式碼審查。這項工具會將待審查的程式碼做為輸入內容 (背景資訊)，並將輸出格式偏好設定做為輸出內容。這項工具會運用大型語言模型分析程式碼，判斷是否正確、有效率、容易維護、安全無虞，以及是否符合最佳做法。這項指令會建構詳細的提示詞，指示 Gemini 如何進行審查，然後傳送至模型來評估提供的程式碼。最後，它會處理 Gemini 的回覆，並根據使用者的偏好設定，將回覆格式化為 Markdown、JSON 或表格，然後輸出評論結果。

法規遵循審查指令：

```
devai review compliance --context ./sample-app/k8s --config ./devai-  
cli/gemini/styleguide.md
```

`devai review compliance` 指令會根據一組最佳做法 (通常是 Kubernetes 設定) 分析程式碼。這項指令會使用 Gemini 模型檢查提供的程式碼 (`context`)，並與另一個設定檔 (`config`) 中定義的標準進行比較。這項指令會利用提示，指示 Gemini 模型扮演 Kubernetes 專家工程師，並提供合規報告。然後將發現項目格式化為簡要說明，重點在於程式碼範例，示範如何解決任何已識別的問題。最後，這項指令會將 Gemini 的法規遵循審查輸出內容列印到控制台。方便使用者輕鬆稽核程式碼是否符合規定。

步驟 9 · 約 0 分鐘

9. 複製 GitLab 存放區

返回 Google Cloud Shell 終端機，然後複製存放區。

```
cd ~  
mkdir gitlab  
cd gitlab
```

請替換為您剛建立的 GitLab 使用者 ID 和存放區網址。

```
git clone https://gitlab.com:YOUR_GITLAB_USERID/genai-for-developers.git
```

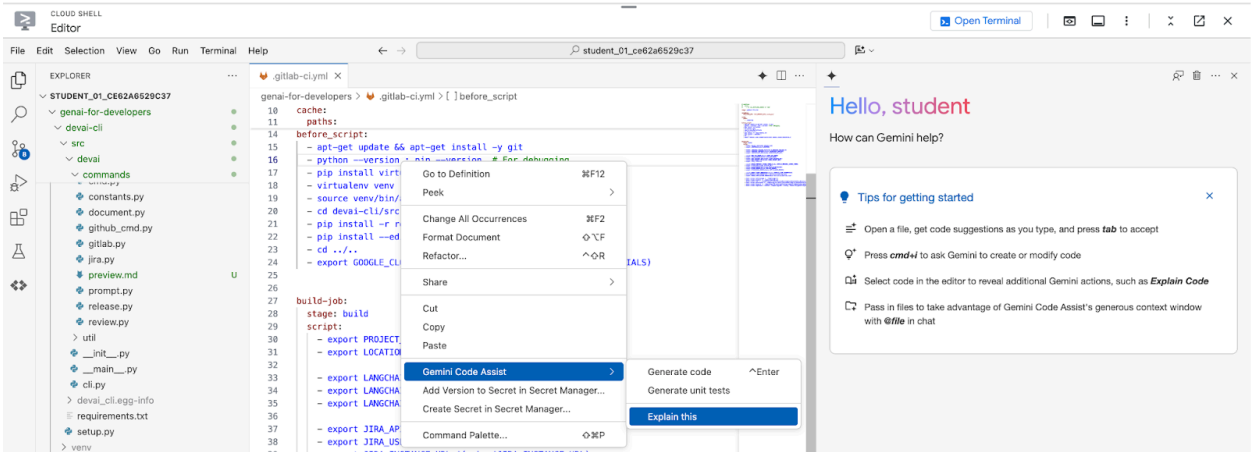
變更目錄並開啟 `.gitlab-ci.yml` 檔案。如果您在匯入期間變更了存放區名稱，請先更新資料夾名稱，再執行下列指令。

```
cd genai-for-developers  
  
cloudshell edit .gitlab-ci.yml
```

步驟 10 · 約 0 分鐘

10. 使用 Gemini Code Assist 解釋程式碼

在 `.gitlab-ci.yml` 檔案的任意處按一下滑鼠右鍵，然後選取 `Gemini Code Assist > Explain`。



審查說明：

Gemini

Okay, I understand. You're asking for an explanation of the `.gitlab-ci.yml` file provided. Let's break down what this file does step-by-step.

Purpose:

The `.gitlab-ci.yml` file is a configuration file used by GitLab's Continuous Integration (CI) system. It defines the automated processes (jobs) that GitLab should run whenever certain events occur in a project's repository, such as code pushes or merge requests.

Structure:

The file is written in YAML, which means it uses indentation and key-value pairs to define its structure. Here's a breakdown of the key components:

- image: python:3.11-slim**
 - image:** This keyword specifies the Docker image that will be used as the execution environment for the CI/CD pipeline.
 - python:3.11-slim:** This is the specific Docker image being used. It's a lightweight image containing Python 3.11, which means that the jobs within this CI/CD pipeline will run in a container with this python version installed.
- variables:**
 - variables:** This section defines environment variables that can be used throughout the pipeline.
 - PIP_CACHE_DIR: "\$CI_PROJECT_DIR/.cache/pip":** This sets the `PIP_CACHE_DIR` variable to a specific directory within the project. This is important for caching Python dependencies installed with `pip`, speeding up subsequent pipeline runs.
- cache:**
 - cache:** This section defines caching policies for the pipeline.
 - paths:** This specifies which directories should be cached.
 - .cache/pip:** The `.cache/pip` directory, where `pip` stores downloaded packages, will be cached between pipeline runs.
- before_script:**
 - before_script:** This defines a set of commands that will run *before* each job in the pipeline. It is like a setup stage.
 - apt-get update && apt-get install -y git:** Updates the package list and installs `git`.
 - python --version && pip --version:** This prints the versions of Python and pip, probably for debugging or informational purposes.
 - pip install virtualenv:** Installs the `virtualenv` package, which allows creating isolated Python environments.
 - virtualenv venv:** Creates a virtual environment named `venv`.
 - source venv/bin/activate:** Activates the `venv` virtual environment.
 - cd devai-cli/src:** Changes the current directory to `devai-cli/src`.
 - pip install -r requirements.txt:** Installs the Python packages listed in the `requirements.txt` file within `devai-cli/src`.
 - pip install --editable .:** Installs the current directory as an editable Python package, which is common during development.
 - cd ../..:** Moves back up two directories.
 - export GOOGLE_CLOUD_CREDENTIALS=\$(echo \$GOOGLE_CLOUD_CREDENTIALS):** Sets the environment variable for google cloud credentials, with the value from the variable that is set on the gitlab ci/cd settings.

11. DevAI CLI 開發

在本節中，您將變更 `devai cli`。

首先，請設定 Python 虛擬環境、安裝必要條件，然後執行範例指令。

```
cd ~/gitlab/genai-for-developers/devai-cli
python3 -m venv venv
. venv/bin/activate
pip3 install -r src/requirements.txt
pip3 install --editable ./src
devai echo
```

設定必要環境變數。

```
export PROJECT_ID=$(gcloud config get-value project)
export LOCATION=us-central1
```

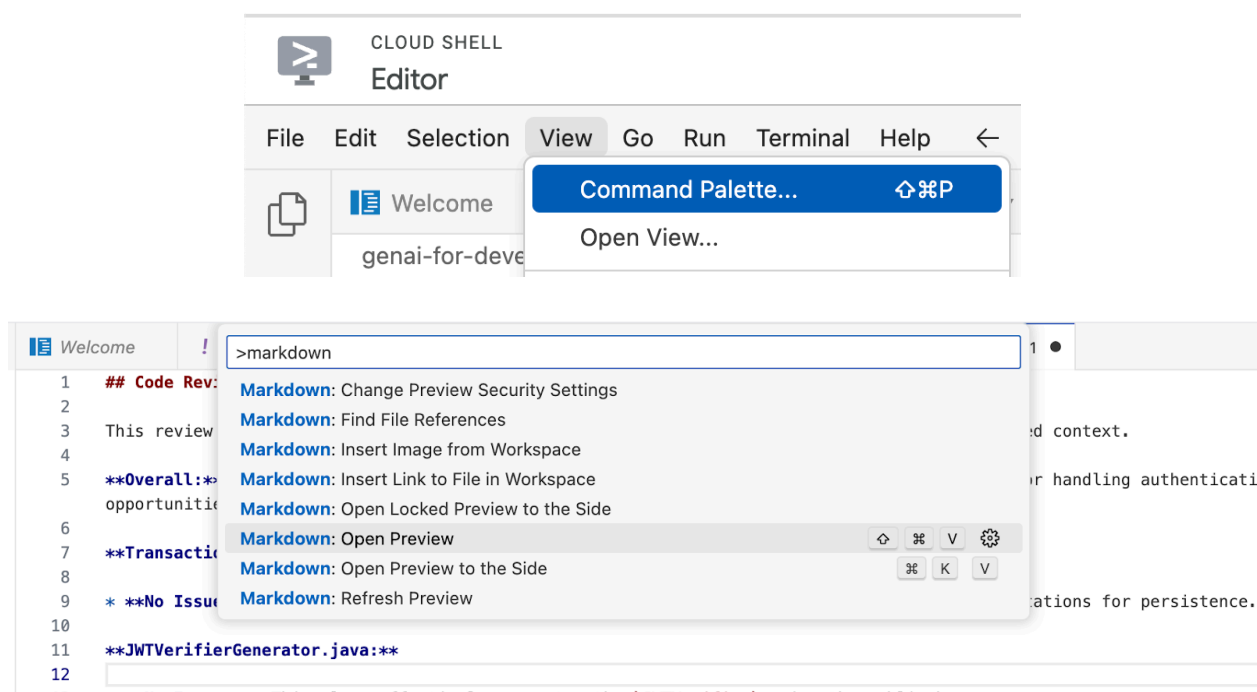
執行程式碼審查指令，確認一切運作正常：

```
devai review code -c ~/gitlab/genai-for-developers/sample-
app/src/main/java/anthos/samples/bankofanthos/balancereader > code-review.md

cloudshell edit code-review.md
```

在 Cloud Shell 編輯器中使用 Markdown 預覽功能查看結果。

使用指令區塊面板，然後選取「Markdown: Open Preview」。



Code Review: BalanceReader Application

This review focuses on the Java code for the BalanceReader application, considering the provided context.

Overall: The code generally appears well-structured and implements common best practices for handling authentication, background tasks, and caching. However, there are opportunities for improvement in error handling, potential race conditions, and code clarity.

Transaction.java:

- No Issues:** The `Transaction` class is a straightforward data model with appropriate annotations for persistence.

JWTVerifierGenerator.java:

- No Issues:** This class effectively generates the `JWTVerifier` using the public key.

BalanceReaderController.java:

- Potential Race Condition (Medium):** The `startWithCallback` method initializes the `latestTransactionId` and starts the background thread. There's a potential race condition where the thread might start processing transactions before the initial `latestTransactionId` is correctly set, potentially leading to missed transactions. Consider using a synchronization mechanism (e.g., a `CountDownLatch`) to ensure the thread starts only after the initial ID is set.
- Error Handling (Medium):** In the `liveness` and `getBalance` methods, the error messages logged are generic (e.g., "Cache error"). Providing more specific error messages, potentially including the underlying exception message, would greatly aid in debugging and monitoring.
- Cache Key (Low):** The cache key is solely the `accountId`. While this might be sufficient in the current context, it could lead to collisions if accounts from different banks (using the same account numbering scheme) are ever used. Consider incorporating the `routingNum` into the cache key for robustness.

TransactionRepository.java:

- No Issues:** The repository interface defines clear queries.

LedgerReader.java:

- Error Handling (Medium):** Similar to `BalanceReaderController`, improve error messages in `startWithCallback` and `getLatestTransactionId` methods for better debugging.
- Poll Frequency (Low):** The `pollMs` value is hardcoded to 100ms. Consider making this configurable via environment variables or configuration files to allow for tuning based on load and desired latency.
- Synchronization (Low):** While not explicitly mentioned, if multiple instances of `LedgerReader` are possible, ensure appropriate synchronization mechanisms (e.g., database-level locking) are in place when reading and updating the `latestTransactionId` to avoid race conditions.

BalanceReaderApplication.java:

- Logging (Low):** The use of `Level.forName("STARTUP", Level.FATAL.intValue())` is unconventional. Consider using standard log levels (e.g., `INFO` or a custom level) for clarity.
- Error Handling (Low):** The application exits immediately if an expected environment variable is not found. While this is acceptable, consider logging the missing variable and gracefully shutting down to allow for monitoring and easier debugging in containerized environments.

BalanceCache.java:

12. 探索 devai CLI 指令

測試涵蓋範圍檢查指令

```
devai review testcoverage -c ~/genai-for-developers/sample-app/src > testcoverage.md  
  
cloudshell edit testcoverage.md
```

法規遵循審查指令

```
devai review compliance --context ~/genai-for-developers/sample-app/k8s --config ~/genai-for-developers/devai-cli/.gemini/styleguide.md > k8s-review.md  
  
cloudshell edit k8s-review.md
```

成效回顧指令

```
devai review performance -c ~/genai-for-developers/sample-app/src/main/java > performance-review.md  
  
cloudshell edit performance-review.md
```

安全審查指令

```
devai review security -c ~/genai-for-developers/sample-app/src/main/java > security-review.md  
  
cloudshell edit security-review.md
```

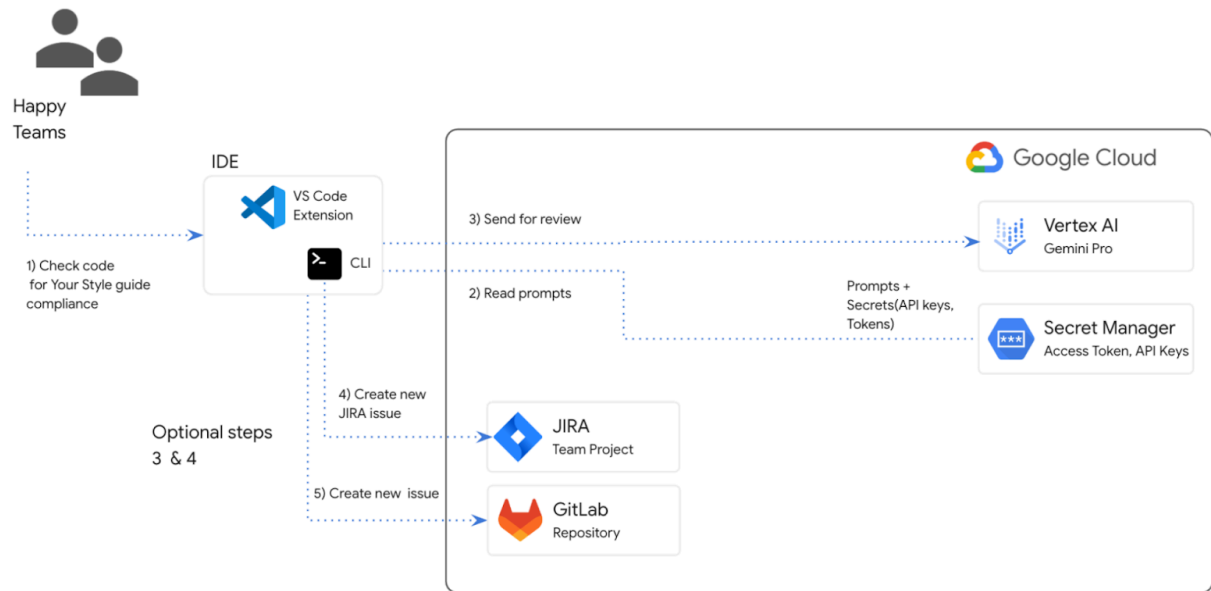
封鎖者審查指令

```
devai review blockers -c ~/gitlab/genai-for-developers/sample-app/pom.xml  
devai review blockers -c ~/gitlab/genai-for-developers/sample-app/setup.md
```

圖片/圖表審查和摘要

輸入圖表 [~/genai-for-developers/images/extension-diagram.png] :

VS Code extension or CLI to perform code review locally



查看指令：

```
devai review image \  
-f ~/genai-for-developers/images/extension-diagram.png \  
-p "Review and summarize this diagram" > image-review.md  
  
cloudshell edit image-review.md
```

輸出內容：

The diagram outlines a process for conducting local code reviews using a VS Code extension or CLI, leveraging Google Cloud's Vertex AI (Gemini Pro) for generating review prompts.

****Process Flow:****

1. ****Code Style Check:**** Developers initiate the process by checking their code for adherence to pre-defined style guidelines.
2. ****Prompt Generation:**** The VS Code extension/CLI sends the code to Vertex AI (Gemini Pro) on Google Cloud.
3. ****Vertex AI Review:**** Vertex AI analyzes the code and generates relevant review prompts.
4. ****Local Review:**** The prompts are sent back to the developer's IDE for their consideration.
5. ****Optional Actions:**** Developers can optionally:
 - Create new JIRA issues directly from the IDE based on the review prompts.
 - Generate new issues in a GitLab repository.

****Key Components:****

- * ****VS Code Extension/CLI:**** Tools facilitating the interaction with Vertex AI and potential integrations with JIRA and GitLab.
- * ****Vertex AI (Gemini Pro):**** Google Cloud's generative AI service responsible for understanding the code and generating meaningful review prompts.
- * ****Google Cloud Secret Manager:**** Securely stores API keys and access tokens required to authenticate and interact with Google Cloud services.
- * ****JIRA/GitLab (Optional):**** Issue tracking and project management tools that can be integrated for a streamlined workflow.

****Benefits:****

- * ****Automated Review Assistance:**** Leveraging AI to generate review prompts saves time and improves the consistency and quality of code reviews.
- * ****Local Development:**** The process empowers developers to conduct reviews locally within their familiar IDE.
- * ****Integration Options:**** The flexibility to integrate with project management tools like JIRA and GitLab streamlines workflow and issue tracking.

圖片差異分析：

```
devai review imgdiff \  
-c ~/genai-for-developers/images/devai-api.png \  
-t ~/genai-for-developers/images/devai-api-slack.png > image-diff-review.md  
  
cloudshell edit image-diff-review.md
```

輸出內容：

The following UI elements are missing in the "AFTER UPGRADE STATE" image compared to the "BEFORE UPGRADE STATE" image:

1. ****Slack:**** The entire Slack element, including the icon, "Team channel" label, and the arrow indicating interaction, is absent in the AFTER UPGRADE image.
2. ****Storage Bucket:**** The "Storage Bucket" element with its icon and "PDFs" label is missing in the AFTER UPGRADE image.
3. ****"GenAI Agents" label in Vertex AI block:**** The BEFORE UPGRADE image has "Vertex AI Agents" and "GenAI Agent" labels within the Vertex AI block, while the AFTER UPGRADE image only has "Vertex AI."
4. ****"Open JIRA Issue" and "Team Project" labels:**** In the BEFORE UPGRADE image, these labels are connected to the JIRA block with an arrow. These are missing in the AFTER UPGRADE image.

****Decision Explanation:****

The analysis is based on a direct visual comparison of the two provided images, noting the presence and absence of specific UI elements and their associated labels. The elements listed above are present in the BEFORE UPGRADE image but absent in the AFTER UPGRADE image.

影片檔案分析：

```
devai review video \  
-f "/tmp/video.mp4" \  
-p "Review user journey video and create unit tests using jest framework"
```

文件生成指令

```
devai document readme -c ~/gitlab/genai-for-developers/sample-app/src/main/
```

輸出內容：


```
# Bank of Anthos - Balance Reader Service
```

Description

The Balance Reader service is a component of the Bank of Anthos sample application. It provides a REST endpoint for retrieving the current balance of a user account. This service demonstrates key concepts for building microservices with Spring Boot and deploying them to a Kubernetes cluster.

Features

- Securely retrieves account balances using JWT authentication.
- Leverages a local cache for fast balance retrieval.
- Asynchronously processes transactions from a central ledger.
- Provides health check endpoints for Kubernetes liveness and readiness probes.
- ...

在 Cloud Shell 編輯器中查看可用的 devai 指令列指令：

```
cloudshell edit ~/gitlab/genai-for-developers/devai-cli/README.md
```

或查看 GitHub 存放區中的 [README.md](#)。

13. 恭喜！

恭喜，您已完成本程式碼研究室！

涵蓋內容：

- 在 GitLab 中新增生成式 AI 程式碼審查自動化步驟
- 在本機執行 [devai](#) CLI

後續步驟：

- 更多實務課程即將推出！

清除所用資源

如要避免系統向您的 Google Cloud 帳戶收取本教學課程所用資源的費用，請刪除含有相關資源的專案，或者保留專案但刪除個別資源。

刪除專案

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